

**Table S1(a): Model GWL Crossing Years and Model Group Assignments (SSP5-8.5)**

|                  | Model      | Scenario | GWL +2C Year    | Winter Model Group (+2C) | Summer Model Group (+2C) | GWL +3C Year    | Winter Model Group (+3C) | Summer Model Group (+3C) |
|------------------|------------|----------|-----------------|--------------------------|--------------------------|-----------------|--------------------------|--------------------------|
|                  | ACCESS-CM2 | SSP5-8.5 | 2038            | High-Frequency           | High-Frequency           | 2056            | Low-Frequency            | Neutral / Other          |
| ACCESS-ESM1-5    | SSP5-8.5   | 2039     | High-Frequency  | High-Frequency           | 2060                     | High-Frequency  | High-Frequency           | High-Frequency           |
| BCC-CSM2-MR      | SSP5-8.5   | 2043     | High-Frequency  | High-Frequency           | 2066                     | High-Frequency  | High-Frequency           | High-Frequency           |
| CESM2            | SSP5-8.5   | 2036     | High-Frequency  | High-Frequency           | 2056                     | High-Frequency  | High-Frequency           | High-Frequency           |
| CESM2-WACCM      | SSP5-8.5   | 2033     | Neutral / Other | Neutral / Other          | 2053                     | High-Frequency  | High-Frequency           | High-Frequency           |
| CMCC-CM2-SR5     | SSP5-8.5   | 2034     | High-Frequency  | High-Frequency           | 2052                     | Low-Frequency   | High-Frequency           | High-Frequency           |
| CMCC-ESM2        | SSP5-8.5   | 2039     | High-Frequency  | High-Frequency           | 2055                     | High-Frequency  | High-Frequency           | High-Frequency           |
| CNRM-CM6-1       | SSP5-8.5   | 2040     | High-Frequency  | High-Frequency           | 2058                     | High-Frequency  | High-Frequency           | High-Frequency           |
| CNRM-CM6-1-HR    | SSP5-8.5   | 2030     | High-Frequency  | High-Frequency           | 2050                     | Neutral / Other | Neutral / Other          | Neutral / Other          |
| CNRM-ESM2-1      | SSP5-8.5   | 2045     | Low-Frequency   | Neutral / Other          | 2064                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| CanESM5          | SSP5-8.5   | 2023     | Low-Frequency   | Low-Frequency            | 2041                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| E3SM-1-0         | SSP5-8.5   | 2032     | Low-Frequency   | Low-Frequency            | 2047                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| EC-Earth3        | SSP5-8.5   | 2036     | Low-Frequency   | Low-Frequency            | 2057                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| EC-Earth3-CC     | SSP5-8.5   | 2035     | Low-Frequency   | Neutral / Other          | 2056                     | Neutral / Other | Low-Frequency            | Low-Frequency            |
| EC-Earth3-Veg-LR | SSP5-8.5   | 2042     | Low-Frequency   | Low-Frequency            | 2061                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| GFDL-ESM4        | SSP5-8.5   | 2053     | Low-Frequency   | Low-Frequency            | 2076                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| HadGEM3-GC31-LL  | SSP5-8.5   | 2030     | High-Frequency  | Low-Frequency            | 2048                     | High-Frequency  | Low-Frequency            | Low-Frequency            |
| HadGEM3-GC31-MM  | SSP5-8.5   | 2034     | High-Frequency  | High-Frequency           | 2050                     | High-Frequency  | Neutral / Other          | Neutral / Other          |
| IITM-ESM         | SSP5-8.5   | 2049     | Neutral / Other | Low-Frequency            | 2076                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| INM-CM4-8        | SSP5-8.5   | 2046     | Low-Frequency   | Low-Frequency            | 2070                     | High-Frequency  | High-Frequency           | High-Frequency           |
| INM-CM5-0        | SSP5-8.5   | 2046     | High-Frequency  | High-Frequency           | 2074                     | High-Frequency  | High-Frequency           | High-Frequency           |
| IPSL-CM6A-LR     | SSP5-8.5   | 2033     | Low-Frequency   | Low-Frequency            | 2051                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| KACE-1-0-G       | SSP5-8.5   | 2024     | Neutral / Other | High-Frequency           | 2043                     | High-Frequency  | High-Frequency           | High-Frequency           |
| KIOST-ESM        | SSP5-8.5   | 2037     | High-Frequency  | High-Frequency           | 2065                     | High-Frequency  | High-Frequency           | High-Frequency           |
| MIROC-ES2L       | SSP5-8.5   | 2047     | Low-Frequency   | Low-Frequency            | 2070                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| MIROC6           | SSP5-8.5   | 2053     | Low-Frequency   | Low-Frequency            | 2076                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| MPI-ESM1-2-HR    | SSP5-8.5   | 2049     | High-Frequency  | High-Frequency           | 2073                     | High-Frequency  | High-Frequency           | High-Frequency           |
| MPI-ESM1-2-LR    | SSP5-8.5   | 2049     | Low-Frequency   | Low-Frequency            | 2072                     | High-Frequency  | High-Frequency           | High-Frequency           |
| MRI-ESM2-0       | SSP5-8.5   | 2038     | Low-Frequency   | Low-Frequency            | 2064                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |
| NorESM2-MM       | SSP5-8.5   | 2054     | High-Frequency  | High-Frequency           | 2076                     | Neutral / Other | High-Frequency           | High-Frequency           |
| UKESM1-0-LL      | SSP5-8.5   | 2032     | Low-Frequency   | Low-Frequency            | 2047                     | Low-Frequency   | Low-Frequency            | Low-Frequency            |